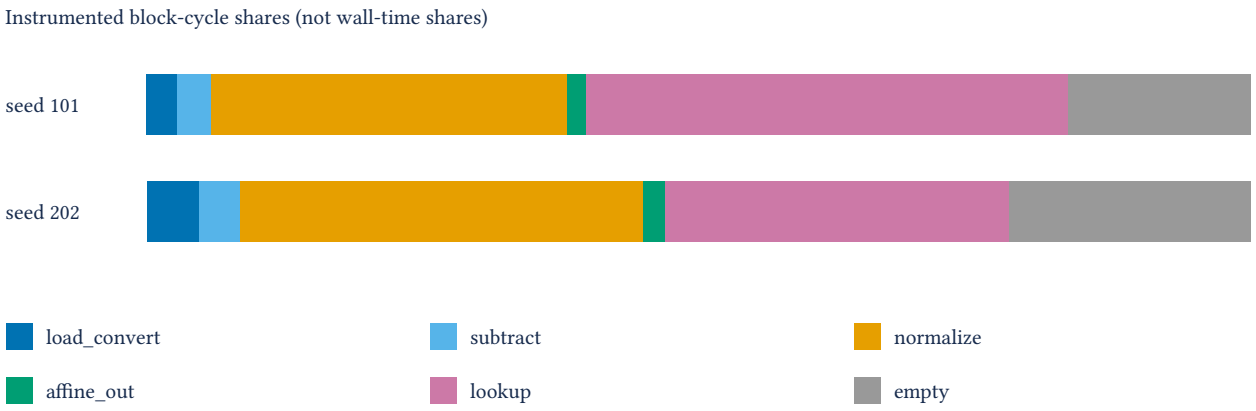
GPU relation-query cost breakdown

RTX 4090 · fixed 63-bit inputs · 31 October 2026

This diagnostic replays the first eight actual queries from each of the two large affine inputs, preserving the original bases, full right cache, filter, and index. Five interleaved rounds yield 30 replay cells and 240 queries. Every response matched the uninstrumented capture byte for byte.



Instrumented block-cycle share	Seed 101	Seed 202
Cache load + input conversion	2.72%	4.72%
Affine subtraction	3.11%	3.72%
Batch normalization (inverse tree)	32.22%	36.55%
Affine output + conversion	1.80%	1.98%
Negative filter + index lookup	43.60%	31.12%
Empty barrier / clock control	16.54%	21.91%

Interpretation. These are sums of thread-zero block-cycle intervals, including barriers, scheduling, and compiler fences. They are not GPU wall-time shares. The empty interval is shown without subtraction. Input conversion includes the cache load; output conversion includes affine reconstruction; lookup includes the negative filter. No fraction is applied to the historical oracle seconds.

Overhead and controls. CUDA events bracket the original or staged search kernel only. Setup, host launch gaps, and winner resolution are excluded. Raw per-round A/A spreads and the staged/stock event ratio are reproduced on the following page. Shared-host clocks remain L0 diagnostics.

Verification and provenance

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Diagnostic prefix: two 63-bit targets, eight queries each, five rounds, three arms. Block
    cycles include barriers and scheduling; NOT additive wall time.
seed=101 queries_per_arm=40 kernel_ms_b=6630.054074999999 kernel_ms_stages=6182.247755
    kernel_ms_x=6617.166291 staged_over_stock=0.933365277272387
    AA_relative_spreads=[0.0010747572281673869, 0.0017998176960455638, 0.0018353295840099568,
    0.0026758852802064277, 0.0023413769175147287]
  load_convert: block_cycles=226842652649 cycles_per_tile=1845.171
    share_including_empty=0.027214
  subtract: block_cycles=259312748053 cycles_per_tile=2109.287 share_including_empty=0.031110
  normalize: block_cycles=2685962164722 cycles_per_tile=21848.004
    share_including_empty=0.322234
  affine_out: block_cycles=149861876447 cycles_per_tile=1218.998
    share_including_empty=0.017979
  lookup: block_cycles=3634373127293 cycles_per_tile=29562.516 share_including_empty=0.436015
  empty: block_cycles=1379074846110 cycles_per_tile=11217.594 share_including_empty=0.165447
seed=202 queries_per_arm=40 kernel_ms_b=3447.530571 kernel_ms_stages=3229.2192993
    kernel_ms_x=3448.417175300001 staged_over_stock=0.936555617328344
    AA_relative_spreads=[0.0002507535540454614, 0.001266927598094058, 0.0001240852263306903,
    0.00047302482779062826, 0.0003653041988215758]
  load_convert: block_cycles=198052529242 cycles_per_tile=2849.916
    share_including_empty=0.047216
  subtract: block_cycles=155936730996 cycles_per_tile=2243.883 share_including_empty=0.037176
  normalize: block_cycles=1533109434379 cycles_per_tile=22060.982
    share_including_empty=0.365498
  affine_out: block_cycles=82883701483 cycles_per_tile=1192.671 share_including_empty=0.019760
  lookup: block_cycles=1305443417065 cycles_per_tile=18784.937 share_including_empty=0.311221
  empty: block_cycles=919155830430 cycles_per_tile=13226.375 share_including_empty=0.219129
```

Validation. All 30 cells / 240 queries passed exact reply, table identity, exit and coverage checks. All 37,584 arithmetic cases passed on host and GPU; 315 differential query cases included 261 hits and 54 exhausted misses. Memcheck, racecheck and synccheck reported zero errors; 29 Rust tests and Clippy passed. Native analysis recomputed byte-identically; canonical export and coverage/file checks reported zero findings. The owned pod was deleted.

Profiler failure. All five optional external profiler invocations exited 2 before collection because the image lacked gawk. No Nsight counters or timeline values were obtained in v1. The separate version-2 protocol repairs prerequisites and attempts only these external profiles; it does not replace this panel.

Perturbation. Staged/stock kernel-event ratios were 0.9334 and 0.9366. The diagnostic kernel specializes the full-cache path and changes code generation as well as adding instrumentation: 58 registers / 56-byte stack versus stock 48 registers / 120-byte stack; neither reports spills. These ratios do not isolate clock overhead or establish a production improvement. The next comparison needs an uninstrumented full-cache specialization on matched complete workloads.

Scope. This is a fixed query-prefix diagnostic, not another complete recovery panel. Instrumentation changes synchronization and register use. The preserved ptxas logs and any supported Nsight counters help assess that perturbation. The source and all data remain reproducible from the linked study artifacts.

Source: ffa0da3efaf222bdcc958d45013aa469d5fbecf7 Protocol: 8b77f5a916607e7f7a0d094d194515d55457823759b9114b97808f7c6416bb17 Study: research/explicit_relations_profile_20261031 Raw data: results-v1/export.json and archive-manifest.json